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Paleomagnetic Investigation of Upper Permian Sediments in the South Junggar Basin, China

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Lacustrine organic-rich mudstones of the Upper Permian Lucaogou Formation are a significant petroleum source bed throughout most of the Junggar basin in northwest China, and paleomagnetic data may provide insight into tectonic conditions prevalent during their deposition. Secondary magnetizations were removed from 180 samples from 29 sites within the South Junggar basin. Characteristic magnetizations of these samples were either revealed directly during thermal demagnetization or were calculated from great circle trajectories. The primary nature of the characteristic magnetization is attested to by a positive fold test, by exclusively reversed magnetizations which are consistent with the mid-Carboniferous to Late Permian Kiaman reversed interval for the geomagnetic field, and by relatively low levels of thermal maturity found in the mudstones. Tectonically corrected remagnetization circles and direct observations combined give a steep direction (I = -61.9°, D = 163.0°, k = 17, n = 78 specimens). The steep mean inclination recorded within the lacustrine organic-rich mudstones suggests that Junggar was situated near its present-day latitude during deposition of the mudstones in Late Permian times, consistent with the marginal ocean basin origin for Junggar that has been proposed based upon stratigraphic relationships. Our data suggest that Junggar may always have been associated with the northern continents of Laurasia, initially as a pre-Permian marginal ocean basin north of the Tethys and subsequently as a Permian, Mesozoic, and Cenozoic nonmarine foreland basin.

Introduction

The potential for significant petroleum reserves in the sedimentary basins of Northwest China has spurred an interest in the geology and tectonics of the region [e.g., *Graham et al.*, 1990] (Figure 1). The Junggar basin first developed as a collisional foreland basin during Permian times, and an Upper Permian lacustrine organic-rich mudstone sequence may be the source for petroleum reservoirs throughout the basin [*Watson et al.*, 1987]. Early Paleozoic ophiolites and Carboniferous deep marine sediments derived from a magmatic arc may ultimately point to an oceanic origin for the Junggar basin [*Feng et al.*, 1989; *Carroll et al.*, 1990]. *Hsü* [1988] proposed a Carboniferous back-arc oceanic setting for Junggar, and he has drawn structural parallels with both the Black Sea and Caspian Sea basins. The Upper Permian lacustrine sediments of Junggar were deposited in a rapidly subsiding, collisional foreland basin, which developed in response to the arrival of the Tarim terrane to the south. Mesozoic and Cenozoic strata in the Junggar basin were deposited in a series of nonmarine foreland basins, under increasingly arid climatic conditions.

Paleomagnetic directions from the late Paleozoic foreland deposits may be helpful in defining both the position of Junggar during the transition from marine to nonmarine facies sedimentation and the motion of Junggar relative to other continents and terranes within and around Pangea. We therefore conducted a paleomagnetic study of Upper Permian andesitic tuffs, nonmarine siltstones, and the lacustrine mudstones exposed along the uplifted southern margin of the Junggar basin. As described below, steep magnetic inclinations recorded in the Permian sequences may place southern Junggar with the bounding calc-alkaline volcanic arc of the northern Tian Shan in high latitudes alongside Laurasia during the Late Permian. These paleomagnetic directions are consistent with an autochthonous history for Junggar, and they lend support to the marginal ocean basin hypothesis.

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Geologic Setting of the Junggar Basin

Despite the occurrence of deformed and dismembered Late Cambrian to Late Devonian ophiolites in fold belts surrounding the Junggar basin [*Feng et al.*, 1989], many workers have suggested that Junggar is underlain by Precambrian crystalline basement. This idea is based entirely upon gravimetrically determined depths to the Moho [e.g., *Ren et al.*, 1987]. It is more likely that Junggar represents a marginal oceanic basin trapped during late Paleozoic subduction and convergence and thus may be entirely underlain by early to mid-Paleozoic oceanic crustal materials [*Carroll et al.*, 1990]. *Kwon et al.* [1989] reported high-level, mid-Carboniferous (321.4 ± 6.7 Ma) anorogenic granites which intrude Paleozoic oceanic sediments and ophiolite mélange exposed in the West Junggar Mountains (Figure 1). From Pb isotope systematics of feldspars, they suggested that the granites were derived from partial melting of oceanic crustal materials, with no contribution from Precambrian basement. Sandstones deposited in the Junggar basin since the Devonian are exclusively of a magmatic arc provenance, further supporting the hypothesis of oceanic crustal materials beneath Junggar [*Carroll et al.*, 1990].

The oldest rocks exposed in southern Junggar appear to be Lower to Middle Carboniferous andesitic volcaniclastics deposited under deep marine conditions [*Carroll et al.*, 1990]. These basinal deposits are in turn overlain by a marine regressive sequence, which consists of carbonate and

volcaniclastic turbidites grading upward into Upper Carboniferous slope and shelf shales, volcaniclastic sandstones, and stromatolitic limestones. The Permian sequence is an accumulation of nonmarine mudstones, siltstones, and fine-grained fluvial sandstones, with the Upper Permian lacustrine organic-rich mudstones constituting a significant 1000 m thick petroleum source bed. These fine-grained lacustrine deposits grade upward into fluvial and alluvial Permo-Triassic conglomerates. Late Permian and later sedimentation was exclusively nonmarine in the Junggar basin, and deposition of over 11 km of sediment in the main depocenter west of Urumqi occurred in a series of foreland basins [*Graham et al.*, 1990; *Carroll et al.*, 1990]. Much of the outcrop relief around Urumqi may be a consequence of compression related to Tertiary convergence between the Indian and Eurasian plates.

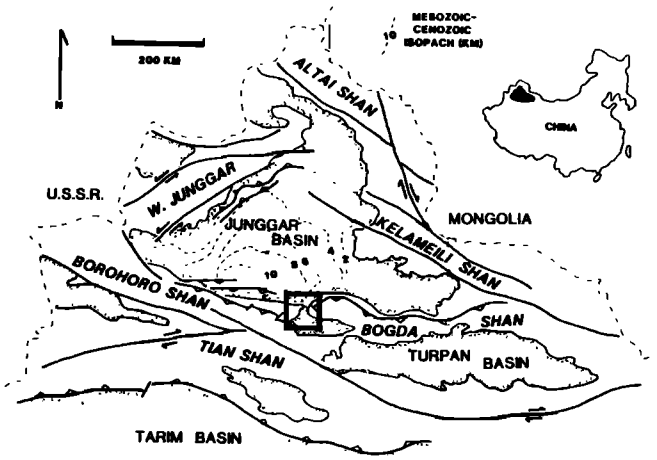


Fig. 1. Location map for northern Xinjiang Province, western China after *Carroll et al.* [1990]. Boxed area indicates the extent of Figure 2.

Andesitic tuffs, siltstones, and lacustrine organic-rich mudstones were collected for paleomagnetic study from the Upper Permian Jingjingzigou, Lucaogou, and Hongyanchi formations (P_2j, P_2l, and P_2h, respectively; listed in ascending stratigraphic order). Each sequence is well exposed in Urumqi at the western end of a large east-west trending, Neogene-faulted anticlinorium found along the northern edge of the Bogda Shan (Bogda Mountains) (Figure 2). The Lucaogou Formation (according to *Graham et al.* [1990]) represents a deep-lake sequence, consists largely of black siliceous mudstone laminated on a millimeter scale, and includes fish remains, phosphatic laminae, and early diagenetic dolostone concretions. The overlying Hongyanchi Formation is a thick sequence of nonlaminated to weakly laminated mudstones with interbedded fluvial sandstones, conglomerates, and carbonate strandline deposits [*Graham et al.*, 1990].

PALEOMAGNETISM

Overview

Linear and planar paleomagnetic directions from the Upper Permian sections near Urumqi on the southern margin of the Junggar basin appear to describe a primary remanence. Three observations support this contention. First, a positive fold test of direct observations of the characteristic magnetization from four sites suggests a predeformation magnetization age. Second, planar demagnetization paths for 129 specimens from 25 sites converge upon a direction statistically indistinguishable from the one defined by the direct observations; such a grouping after correction for tilt of beds is

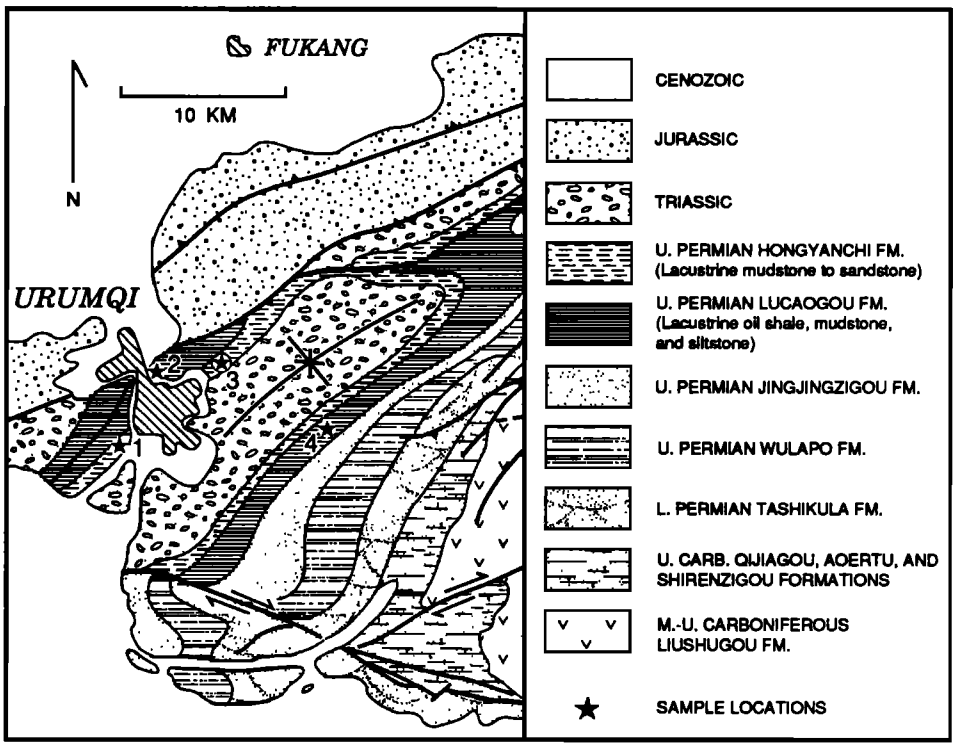


Fig. 2. Generalized geologic map and stratigraphy of the Urumqi region [after *Bureau of Geology and Mineral Resources of Xinjiang Uygur Autonomous Region*, 1983]. The locations of the sections are as follows: 1, Quarry; 2, Hongshan; 3, Cemetery; and 4, East. Abbreviations in the legend are as follows: FM., Formation; U., Upper; L., Lower; and M.-U., Middle to Upper. Bold lines mark fault traces, and those with sawteeth signify thrust faulting.

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also a positive fold test. Third, the occurrence of only reversed magnetizations is consistent with the mid-Carboniferous to Late Permian Kiaman reversed interval for the geomagnetic field [*Harland et al.*, 1982]. After tectonic correction, a mean inclination of -62° suggests no net latitudinal motion for Junggar since Permian times. Similar magnetic inclinations from other late Paleozoic sections sampled both in the Bogda Shan east of Urumqi and in the West Junggar Mountains across the basin attest to the structural integrity of the Junggar "block" through Mesozoic and Cenozoic times [*Li et al.*, 1989; *Li et al.*, 1991].

Sampling

Upper Permian samples were collected from 34 sites at four separate localities in and east of the city of Urumqi (Figure 2). Six individually cored and oriented samples on average from a distinct stratigraphic time horizon constitute a site. Four sites at the East Section locality were collected from the Jingjingzigou Formation; 11 sites at the Hongshan Section locality, 10 sites at the Cemetery Section locality, and an additional four sites at the East Section locality were all collected from the Lucaogou Formation; and five sites from the Quarry Section locality were collected from the Hongyanchi Formation. Both limbs of an anticline were sampled at the Hongshan Section locality. Both limbs of a syncline were sampled at the Cemetery Section. The Quarry and East sections are both monoclinally dipping. The azimuthal orientation of each core consisted of a magnetic reading for all 215 samples, with a Sun azimuth for 122 samples.

Specimens were cut from each cylindrical sample, and the specimen farthest from the original outcrop surface was labeled A. A second specimen, if available, was labeled B. Standard magnetic measurement and demagnetization procedures were conducted at Stanford University, Stanford, California, using an automated cryogenic magnetometer with integrated alternating field (AF) demagnetization coils, and in air in a Schonstedt TSD-1 oven. Bulk magnetic susceptibilities were measured with a Bartington MS2 magnetic susceptibility meter.

Demagnetization

Most A specimens from the 34 sites collected were thermally demagnetized. Each thermally demagnetized specimen was cooled in a residual field not greater than 3 nanotesla (nT). To remove any viscous component of magnetization acquired while transferring the specimen to the magnetometer from the oven, the specimens were subjected to an alternating field of 10 millitesla (mT) within the low-field space of the magnetometer just prior to measurement.

Pilot studies. To provide a preliminary assessment of the magnetization of the Lucaogou and Hongyanchi formations, three A specimens were demagnetized thermally, while the three corresponding B specimens were demagnetized using alternating fields. Each A specimen was heated up to 500°C in 13 steps. In parallel experiments the B specimens were demagnetized in increasing alternating fields with peak values from 5 to 80 mT. Alternating field demagnetization in each case defined just one component of magnetization identical to the present-day field direction at Urumqi ($D = 3°$, $I = 63°$) (see specimens 8UP22.2.B, 8UP22.5.B, and 8UP31.1.B in Figure 3). Demagnetization, however, was not complete, as a significant proportion of the natural remanent magnetization

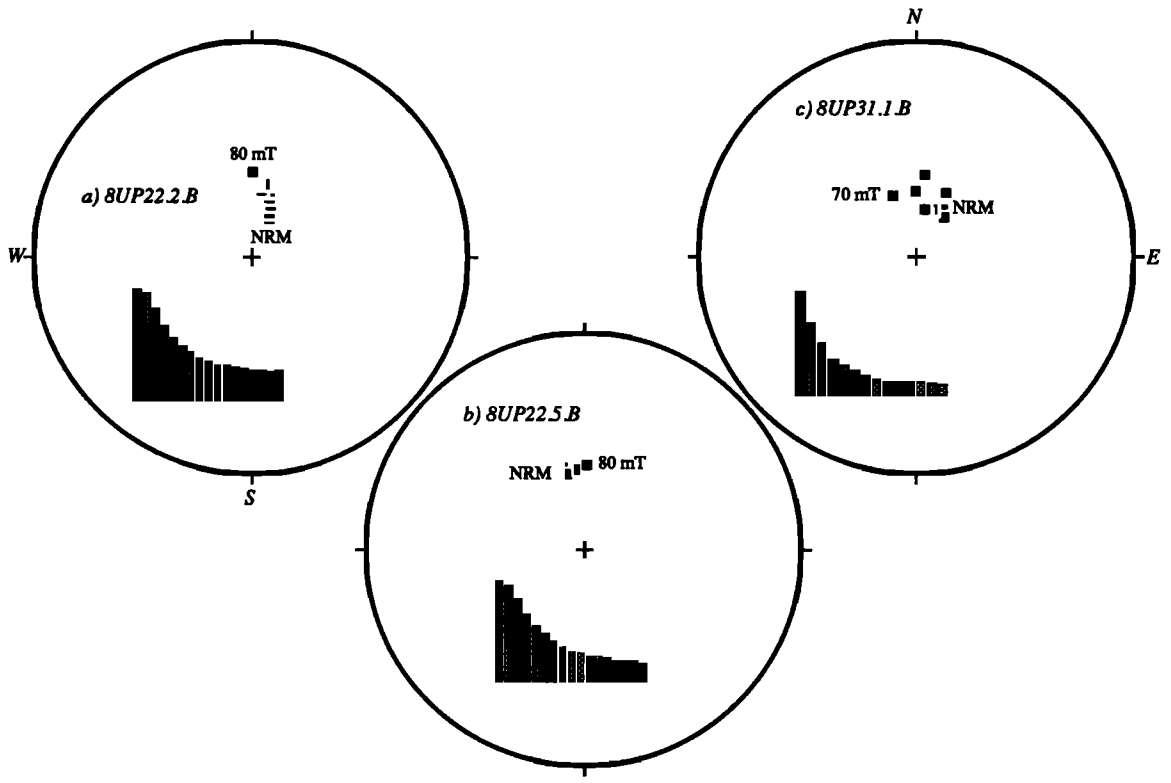


Fig. 3. Equal-area stereographic projections of alternating field pilot demagnetization for specimens (*a*) 8UP22.2.B, (*b*) 8UP22.5.B, and (*c*) 8UP31.1.B. Solid symbols on the stereographic plot indicate projections onto the lower hemisphere. Inset bar graphs plot normalized intensity of magnetization versus intensity of demagnetizing field. All projections are of in situ directions.

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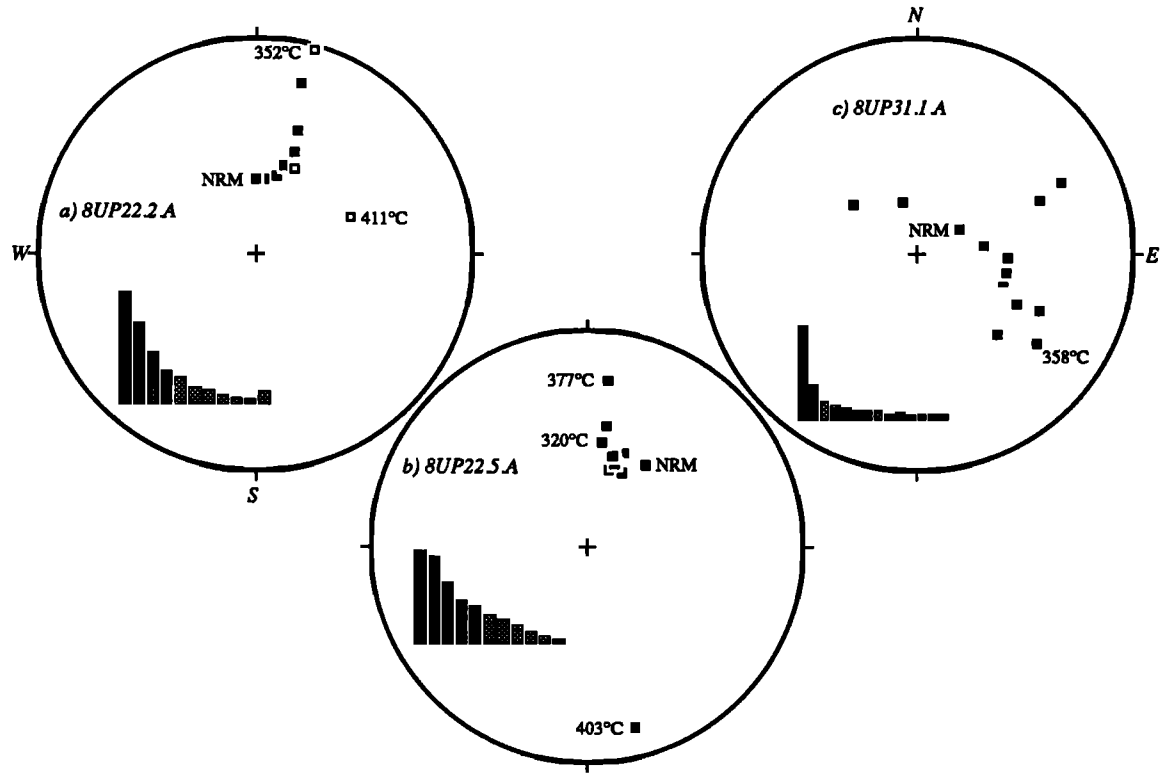


Fig. 4. Equal-area stereographic projections of thermal pilot demagnetization for specimens (*a*) 8UP22.2.A, (*b*) 8UP22.5.A, and (*c*) 8UP31.1.A. Solid symbols on the stereographic plot indicate projections onto the lower hemisphere; open symbols indicate projections onto the upper hemisphere. Inset bar graphs plot normalized intensity of magnetization versus demagnetizing temperature. All projections are of in situ directions.

(NRM) remained after treatment in alternating fields of 80 mT. Thermal demagnetization behavior of the A specimens was different from, yet consistent with, the AF results (Figure 4). In each of the three A cases, the same present-day component of magnetization was removed, but more efficiently and completely. After heating to 250°C and greater, the direction of remanent magnetization for the three swung away from the present-day direction along what appear to be great circle paths toward unknown directions. Organic material within the oil shales (total organic carbon values range from 0.84 to 34.27% [*Graham et al.*, 1990]) began to pyrolyze at temperatures of 425°C, precluding the possibility of directly observing the "underlying" components of magnetization.

The intensity of the present-day component of magnetization must be significantly larger than the intensity of the underlying component, since the NRM direction is essentially that of the present-day component. Since AF demagnetization removes only part of the present-day component of magnetization, the coercive force (H_c) of the present-day component must be greater than 80 mT. In contrast to coercivity, unblocking temperatures (T_{ub}) of the present-day component of magnetization are much less than temperatures for the underlying component. As the present-day component of magnetization demagnetizes with increasing temperature, the remanent magnetization direction moves from the present-day direction toward the underlying direction (Figure 4). An iron hydroxide such as goethite (α-FeOOH) could be carrying the high-H_c/low-T_{ub} component of magnetization [*Dekkers*, 1989*a*, *b*], while magnetite with specific grain size and shape distributions could account for the high-T_{ub} underlying component.

Jingjingzigou Formation sediments and tuff at the East Section. Specimens from three of the four East Section sites gave direct observations of a high-T_{ub} reversed component of magnetization in response to thermal demagnetization. A low-T_{ub} present-day direction and the high-T_{ub} reversed direction were the two components of coherent magnetization resolved in siltstone specimens from sites 8UP33, 8UP34, and 8UP35 (Figure 5). Each thermally demagnetized specimen was heated 15 times from 175° to 600°C in combinations of 50° and 25°C increments. Intensities of NRM ranged from 1×10^{-3} to $6 \times$

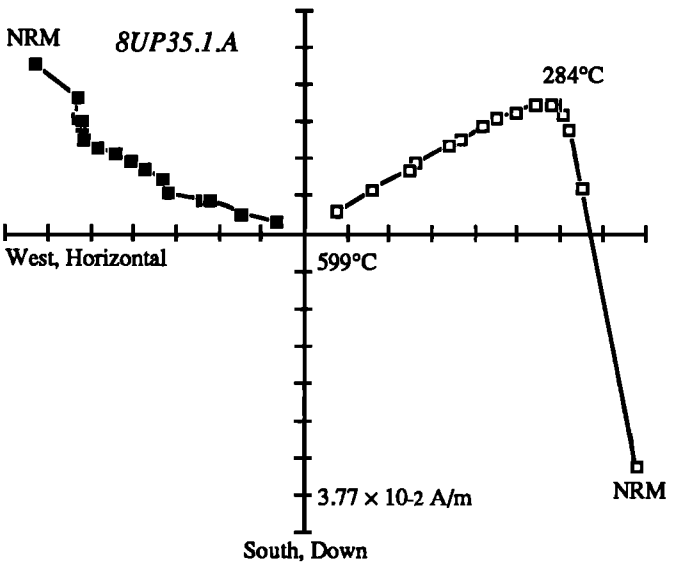


Fig. 5. Orthogonal vector plot of thermal demagnetization for specimen 8UP35.1.A. Solid symbols on the orthogonal plot indicate projections onto the horizontal plane; open symbols indicate projections onto a vertical plane. All projections are of in situ directions.

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10^{-2} ampere/meter (A/m). A random, postcollection component of magnetization was unblocked completely after heating to 75°C. The present-day component of magnetization was removed after heating to 300°C, isolating the reversed magnetization which decayed univectorially toward zero during additional heating to 575°C. The final intensity of magnetization for each specimen was at least an order of magnitude less than the NRM value.

Specimen 8UP35.7.A was demagnetized by AF means, which resolved the same two components of magnetization (Figure 6). The reversed direction of magnetization was isolated in alternating fields with peak values of 25 mT. Coercivities of the reversed component were in excess of 70 mT.

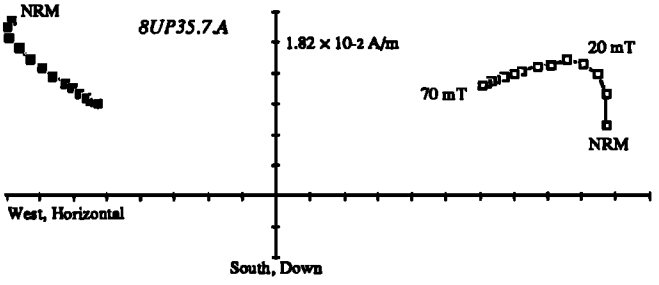


Fig. 6. Orthogonal vector plot of alternating field demagnetization for specimen 8UP35.7.A. Solid symbols on the orthogonal plot indicate projections onto the horizontal plane; open symbols indicate projections onto a vertical plane. All projections are of in situ directions.

Site 8UP32 was collected stratigraphically high in the Jingjingzigou Formation, within a conchoidally fractured, well-indurated andesitic tuff layer. Each of the six specimens was heated 13 times from 75° to 525°C in combinations of 75°, 50° and 25°C steps. A random, postcollection direction and most of the present-day component of magnetization were removed during heating in temperatures of 270°C and less. Additional heating steps to 525°C moved the direction of remanent magnetization toward the northwesterly declination and negative inclination direction resolved in specimens from the other three sites (Figure 7), but linear decay to the origin was never really observed. Intensities of NRM were of the order of 3×10^{-4} A/m; final intensities averaged 9×10^{-5} A/m.

Hongyanchi Formation sediments at the Quarry Section. Samples were collected from mudstone and siltstone horizons at five sites in the Hongyanchi Formation exposed at the southeast dipping Quarry Section (sites 8UP27-8UP31, Figure 2). Of all the Hongyanchi Formation specimens, the six from site 8UP28 were unique in that direct observations of the high-T_{ub} component of magnetization were made (Figure 8). Thermal demagnetization revealed two coherent components of magnetization: the same low-T_{ub} present-day component, and the higher-T_{ub} component with an in situ inclination of -8.8° and declination of 127.8° (k = 34, a_{95} = 11.7°, n = 6 specimens). Intensities of NRM for the site 8UP28 specimens averaged 4×10^{-4} A/m. A random, postcollection component of magnetization was destroyed after heating to 75°C; the present-day component unblocked completely after heating to 280°C, and the reversed, characteristic component was well delineated as it decayed toward zero with subsequent heating to 490°C. Final intensities of magnetization dropped an order of magnitude from NRM values to an average of 4×10^{-5} A/m.

The remaining A specimens from the Quarry Section sites, except for two from site 8UP31, demagnetized partially with

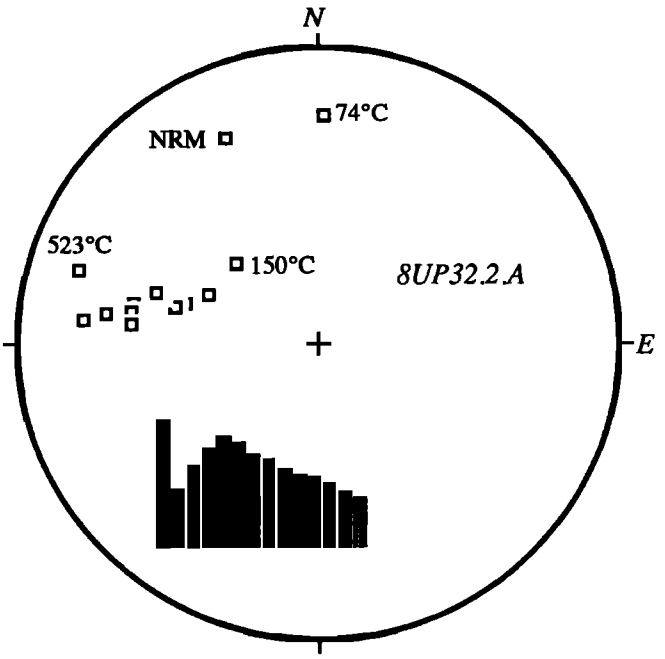


Fig. 7. Equal-area stereographic projection of thermal demagnetization for specimen 8UP32.2.A. Open symbols on the stereographic plot indicate projections onto the upper hemisphere. Inset bar graph plots normalized intensity of magnetization versus demagnetizing temperature. All projections are of in situ directions.

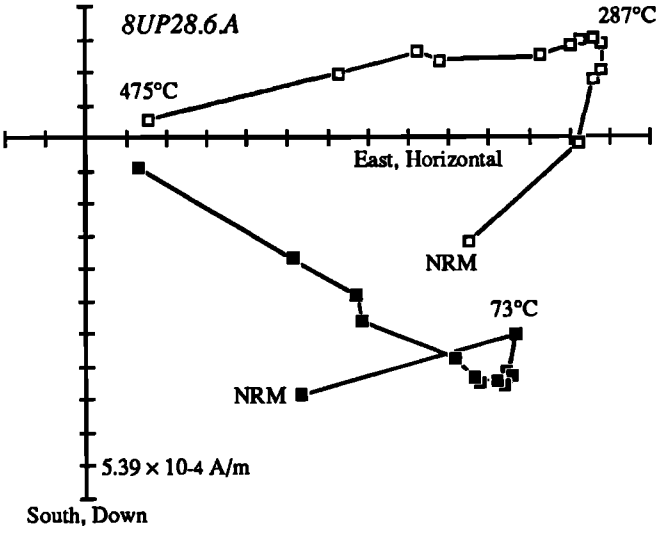


Fig. 8. Orthogonal vector plot of thermal demagnetization for specimen 8UP28.6.A. Solid symbols on the orthogonal plot indicate projections onto the horizontal plane; open symbols indicate projections onto a vertical plane. All projections are of in situ directions.

the remanent magnetization moving away from the NRM present-day direction toward the characteristic direction isolated with site 8UP28. Natural remanent magnetization intensities averaged 5×10^{-4} A/m. Final intensities of magnetization averaged 3×10^{-5} A/m after heating to temperatures of 486°C.

Lucaogou Formation sediments by location. Samples were collected from mudstone and siltstone horizons at 25 sites in the Lucaogou Formation, which was exposed at three sampling localities. Sites 8UP13-8UP22 came from both limbs of a northeast trending syncline at the Cemetery Section. Sites 8UP23-8UP26 were collected from the northwest dipping East

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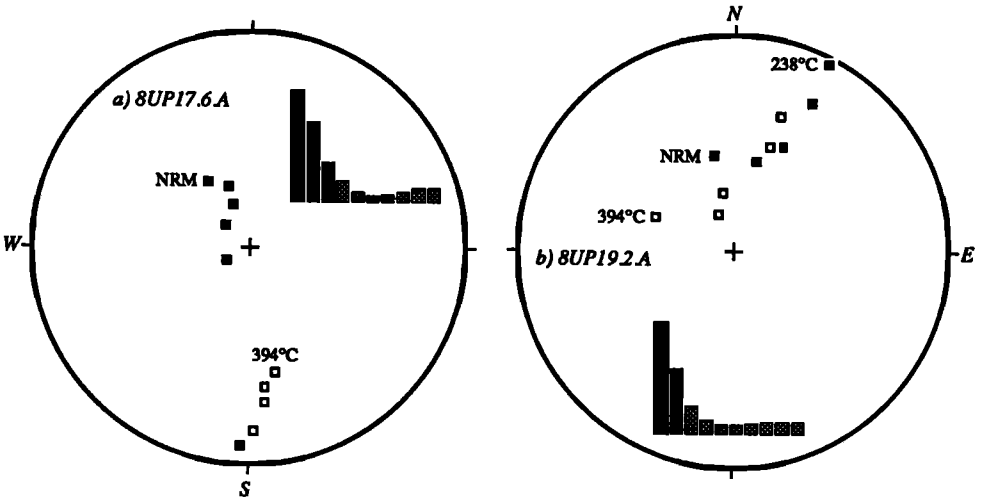


Fig. 9. Equal-area stereographic projections of thermal demagnetization for Cemetery Section specimens (*a*) 8UP17.6.A and (*b*) 8UP19.2.A. Solid symbols on the stereographic plot indicate projections onto the lower hemisphere; open symbols indicate projections onto the upper hemisphere. Inset bar graphs plot normalized intensity of magnetization versus demagnetizing temperature. All projections are of in situ directions.

Section, and sites 8UP01-8UP11 came from both limbs of the Hongshan anticline in Urumqi (Figure 2). From bedding attitude measurements, the anticline appears to plunge (unlike the Cemetery Section syncline) toward the northeast. Following the pilot demagnetization results, 120 A specimens from 20 Lucaogou Formation sites were heated 8-15 times from 75°C to in a few cases 600°C in combinations of 100°, 75°, 50° and 25°C increments. Heating at temperatures of 425°C and higher caused both the occurrence of combustion within most of the mudstone samples as organic material rapidly oxidized and an increase in bulk magnetic susceptibilities.

Specimen directions from the Cemetery Section (except for 11) all followed great circle trajectories during thermal demagnetization, swinging away from a present-day field direction toward a high-T_{ub} direction. The direction toward which the remanent magnetization moved depended upon the structural setting, suggesting dispersion was introduced by tectonic folding and perhaps faulting (Figure 9). Intensities of NRM for the specimens from the 10 sites measured (sites 8UP13-8UP22) ranged almost an order of magnitude between 4 $\times 10^{-4}$ and 3 $\times 10^{-3}$ A/m and averaged 1 $\times 10^{-3}$ A/m. An inconsistent, probably postcollection component of magnetization (if present) was removed with heating before 75°C (Figure 9). With increasing temperature, the remanent magnetization of almost all specimens changed in direction and intensity as the present-day component was removed. Specimen 8UP17.6.A from the southeast dipping, northwest limb of the syncline typifies this great circle behavior, as its remanent magnetization leaves the present-day direction almost immediately upon heating and moves toward a direction with an in situ southerly declination and a negative inclination (Figure 9). Specimen 8UP19.2.A from the northwest dipping, southeast limb of the syncline displays identical behavior; however, the direction pursued has an in situ steep negative inclination and a northwesterly declination (Figure 9). Final intensities of magnetization averaged 1 $\times 10^{-4}$ A/m at 400°C.

Thermal demagnetization of 11 specimens failed to reveal any magnetization other than the present-day component, which in these specimens may be substantially larger than any other. These 11 were removed from further consideration and analysis.

With the exception of six specimens that revealed only a present-day field direction, the northwest dipping East Section specimens (sites 8UP23-8UP26) displayed great circle trajectories during thermal demagnetization (Figure 10). The direction toward which the remanent magnetizations moved had in situ shallow negative inclinations and north-northwesterly declinations. Natural remanent magnetization intensities averaged 3 $\times 10^{-4}$ A/m and fell to 30% of that value with thermal demagnetization to 370°C.

Most of the Hongshan specimens (sites 8UP01-8UP11) again showed great circle remanent magnetization paths starting from the present-day field direction and ending near a

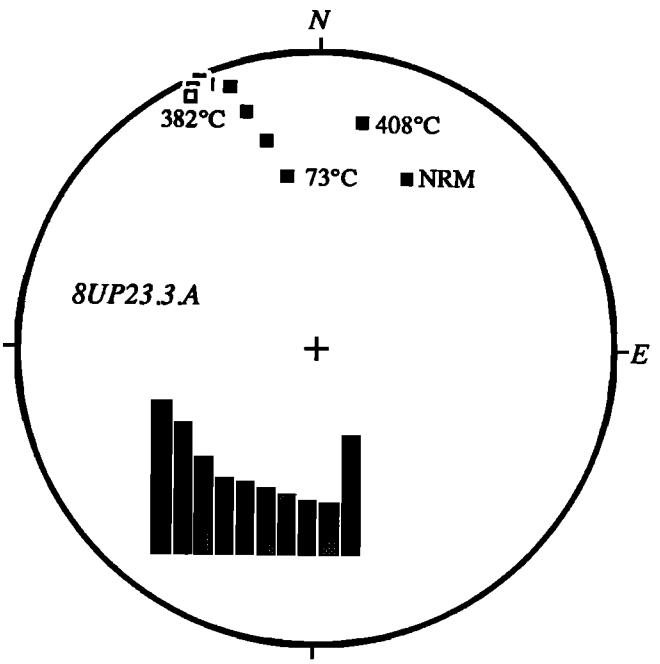


Fig. 10. Equal-area stereographic projection of thermal demagnetization for an East Section specimen. Solid symbols on the stereographic plot indicate projections onto the lower hemisphere; open symbols indicate projections onto the upper hemisphere. Inset bar graph plots normalized intensity of magnetization versus demagnetizing temperature. All projections are of in situ directions.

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high-T_{ub} reversed direction (Figure 11). The amount of great circle distance traveled varied from 6° to 171°. An average NRM intensity of 2×10^{-3} A/m decreased monotonously to 1×10^{-4} A/m at temperatures of 415°C. A small, postcollection, random component of magnetization was removed immediately with heating, and subsequent unblocking of the present-day component defined a great circle path beyond which the characteristic component lay.

Calculation of Mean Directions

The multivariate technique of principal component analysis described by *Kirschvink* [1980] was used to find the best least squares line or plane along the demagnetization path for each specimen. The best least squares high-T_{ub} direction (line) of magnetization for the Jingjingzigou Formation specimens with direct observations (sites 8UP33-8UP35) was determined by an average of nine magnetic vectors during thermal demagnetization between 346° and 553°C. For the six specimens from the Hongyanchi Formation (site 8UP28), the directly observed characteristic direction of magnetization was best defined by five magnetic vectors following thermal demagnetization between 346° and 474°C. The best fit for the one AF demagnetized, direct observation specimen (8UP35.7.A) included the origin plus 10 additional points defined between 25 and 70 mT. With the exception of one other, thermally demagnetized specimen, all least squares linear segment fits to observed high-T_{ub} directions of magnetization were not anchored to the origin. Site mean directions are listed in Table 1.

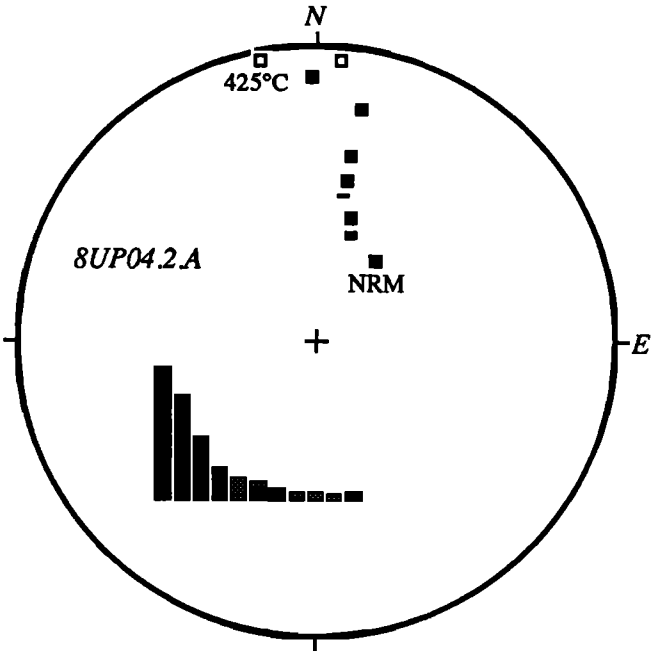


Fig. 11. Equal-area stereographic projection of thermal demagnetization for a Hongshan Section specimen. Solid symbols on the stereographic plot indicate projections onto the lower hemisphere; open symbols indicate projections onto the upper hemisphere. Inset bar graph plots normalized intensity of magnetization versus demagnetizing temperature. All projections are of in situ directions.

Aberrant points were first edited from the thermal demagnetization paths of the remaining specimens before least squares planar fits were attempted. Each demagnetization plane was anchored to the origin, assuming a contribution of only two components of magnetization to the vector remanent magnetization sum. The pole to a demagnetization plane is an estimate of where the two defining magnetic components do not lie. In situ demagnetization paths diverge from the present-day field direction of magnetization; tectonically corrected demagnetization paths should converge upon the second component. Poles to great circle demagnetization planes for 129 specimens are listed in Table 2.

Least squares linear fits of thermal demagnetization steps less than 250°C defined the low-T_{ub} present-day component of magnetization. The in situ mean direction for the present-day components of magnetization from the 29 measured sites (D = 5.0°, I = 63.5°, k = 113, a_{95} = 2.5°, n = 29 sites) is indistinguishable from the dipole field direction (D = 0.0°, I = 62.5°) at Urumqi.

Results

An opportunity for the application of *Graham*'s [1949] fold test is provided by the four sites at which direct observations were made (Figure 12). If directions of magnetization are brought into agreement after bedding planes are restored to horizontal about lines of strike, it can be argued that the magnetizations have remained unchanged since the time of deformation. Using *McElhinny*'s [1964] statistical procedure to evaluate overall improvement in the grouping of the high-T_{ub} directions, a fold test (k_2/k_1 = 20.85, n = 4 sites) is

TABLE 1. Compilation of Direct Observation Paleomagnetic Data

Site	D	I	D_c	I_c	S	Dip	N	R	k	a_{95}
Quarry Section P_2h (Latitude, 43.77°N; Longitude, 87.58°E)										
UP28	127.8	-8.8	130.8	-59.4	34.6	50.7SE	6	5.8528	34	11.7
East Section P_2j (Latitude, 43.78°N; Longitude, 87.75°E)										
UP33	313.7	-26.9	184.6	-75.0	234.5	74.3NW	7	6.9896	576	2.5
UP34	305.0	-26.6	157.9	-62.8	226.0	88.5NW	6	5.9988	4308	1.0
UP35	297.6	-33.4	173.6	-54.1	230.3	85.6NW	7	6.8878	53	8.3
Mean			159.3	-64.2			4	3.9174	36	15.5

D is in situ declination in degrees; I is in situ inclination in degrees; D_c is corrected declination in degrees; I_c is corrected inclination in degrees; S is strike in degrees; Dip is dip in degrees and dip direction; N is number of specimens; R is resultant vector; k is precision parameter estimate; a_{95} is radius of 95% confidence in degrees. Each site is listed separately with location, and a tilt-corrected mean (N = 4 sites) is given.

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TABLE 2. Compilation of Remagnetization Circle Paleomagnetic Data

Specimen	A	P	A_c	P_c	Arc	N	Range	MAD	VRM_D	VRM_I
Cemetery Section (Latitude, 43.82°N; Longitude, 87.65°E)										
Site UP18 P_2l (S = 63.8°, D = 82.0°S)										
1.A									6.2	36.9
2.A	52.4	-0.5	61.7	11.2	36.0	4	0-253	3.1	323.2	68.0
3.A	268.8	-1.4	249.0	24.5	33.7	9	98-371	8.5	356.1	64.5
4.A	269.8	-4.0	252.0	25.0	32.6	7	242-369	13.7	34.2	56.4
5.A	273.1	-1.6	250.1	28.7	35.9	10	83-387	10.6	2.7	65.1
6.A	237.6	1.3	241.6	-6.0	44.5	11	0-391	13.7	346.9	72.9
7.A									356.1	72.1
Site UP17 P_2l (S = 59.5°, D = 77.0°S)										
1.A	242.4	4.5	235.8	3.8	13.8	6	312-449	8.1	335.9	61.2
2.A	279.3	1.1	248.8	38.9	82.3	5	343-445	7.2	349.7	66.9
3.A	277.3	-17.5	268.9	30.1	145.7	10	83-387	8.8	340.0	63.0
4.A	239.4	15.8	224.1	3.4	131.5	9	167-391	10.9	1.9	64.8
5.A	267.6	-17.6	264.7	21.7	31.0	5	284-393	5.6	351.7	60.8
6.A	268.8	-9.2	256.6	25.8	139.7	8	216-394	5.5	355.3	65.4
Site UP16 P_2l (S = 57.5°, D = 71.0°S)										
1.A	265.3	5.7	241.2	28.1	50.2	5	336-441	11.6	354.2	62.8
2.A	250.1	11.0	230.9	15.3	87.4	11	86-444	7.6	357.7	59.6
3.A	257.5	4.0	240.3	20.2	16.5	5	284-393	8.6	2.1	74.1
4.A	246.8	9.5	231.4	11.8	7.5	5	301-394	6.6	6.2	71.5
5.A	245.4	10.1	230.4	10.7	12.2	9	184-396	11.1	8.1	68.5
6.A	256.2	5.4	238.4	19.4	41.7	11	96-450	6.9	0.7	59.6
Site UP15 P_2l (S = 60.0°, D = 75.0°S)										
1.A	248.6	8.1	234.3	10.3	18.2	11	0-363	13.7	357.5	66.2
2.A	261.1	6.8	238.6	22.1	58.5	12	72-398	18.0	11.3	70.8
3.A	244.5	8.2	233.2	6.4	42.2	11	72-415	17.9	357.7	69.8
4.A	248.8	7.1	235.3	10.3	62.0	11	72-398	14.2	1.3	72.0
5.A	246.7	7.9	234.0	8.5	24.8	10	72-388	14.0	3.4	72.9
6.A	239.7	9.2	231.0	2.1	23.9	8	196-373	16.0	19.6	77.1
Site UP14 P_2l (S = 59.5°, D = 73.0°S)										
1.A	254.4	6.5	237.5	16.1	59.9	7	0-385	3.3	355.5	49.5
2.A	267.9	8.0	239.8	29.4	35.9	5	99-325	2.5	10.6	62.6
3.A	254.5	3.7	240.3	15.4	49.1	9	99-408	4.3	353.2	70.7
4.A	255.8	4.8	239.6	17.0	54.6	8	98-380	5.0	353.0	66.3
5.A	265.8	0.9	246.8	25.4	45.8	8	98-377	4.1	359.7	72.1
6.A	256.4	5.4	239.2	17.7	50.1	9	98-398	4.3	1.7	65.4
Site UP13 P_2l (S = 61.1°, D = 73.5°S)										
1.A	254.6	-1.3	246.3	12.6	48.9	7	100-385	3.5	347.9	73.2
2.A	246.5	-2.8	245.3	4.4	48.2	9	99-411	11.5	290.9	73.1
3.A	255.3	13.3	231.9	17.1	45.9	8	99-382	3.6	16.4	62.3
4.A	249.3	6.3	237.3	9.6	47.8	8	99-380	6.8	352.2	66.6
5.A	249.1	6.4	237.2	9.5	52.4	9	98-403	3.8	356.1	73.4
6.A	257.7	11.7	234.1	19.0	59.5	10	98-426	6.1	3.6	55.3
Site UP22 P_2l (S = 227.9°, D = 49.0°NW)										
1.A	50.9	-20.1	65.2	-10.9	33.9	5	269-354	10.9	15.8	70.7
2.A	104.6	-5.3	96.3	34.6	51.8	5	267-352	3.4	39.0	74.4
3.A	80.8	-8.8	77.3	17.7	99.9	7	241-382	6.4	6.8	63.9
4.A	49.4	-19.4	63.7	-11.5	33.4	6	265-380	10.0	22.0	68.2
5.A	97.7	-0.4	86.0	34.9	26.2	4	320-377	3.2	8.5	64.9
6.A	21.4	-31.4	58.5	-39.0	27.2	5	287-372	6.7	29.8	58.3
7.A	59.2	-23.7	73.0	-7.4	33.7	6	256-363	5.2	28.0	62.7
8.A	71.2	-20.2	78.2	3.1	29.2	4	297-348	0.3	23.3	61.3
Site UP21 P_2l (S = 230.3°, D = 47.5°NW)										
1.A	104.6	-6.3	97.5	31.4	24.0	4	281-363	6.5	9.8	61.3
2.A									17.5	62.8
3.A									7.9	61.1
4.A									14.2	53.3
5.A	118.8	11.7	102.7	54.0	43.2	5	97-281	12.5	353.1	66.2
6.A	103.8	12.5	82.8	46.5	28.3	5	271-348	7.7	7.5	48.4
Site UP20 P_2l (S = 227.4°, D = 40.0°NW)										
1.A									12.0	57.5
2.A									16.8	55.9
3.A									39.0	65.9
4.A									6.2	61.2
5.A									24.9	60.1
6.A									46.7	41.4
Site UP19 P_2l (S = 222.5°, D = 39.5°NW)										
1.A	127.9	-2.7	126.8	36.7	18.8	6	256-393	4.8	54.3	59.0
2.A	117.1	-16.3	116.6	21.8	112.3	8	216-394	23.9	2.9	59.9

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TABLE 2. (continued)

Specimen	A	P	A_c	P_c	Arc	N	Range	MAD	VRM_D	VRM_I
3.A	103.5	-20.1	104.4	14.9	130.6	8	224-396	14.6	50.3	61.5
4.A	207.4	24.6	227.8	28.2	119.4	9	96-371	29.2	8.2	66.6
5.A	125.2	-5.9	123.8	33.2	145.8	8	237-396	24.2	26.2	67.9
6.A	102.2	-9.9	99.5	24.1	17.7	5	242-315	9.8	6.2	67.3
East Section (Latitude, 43.78°N; Longitude, 87.75°E)										
Site UP26 P_2l (S = 239.8°, D = 89.0°NW)										
1.A									354.3	68.3
2.A	91.9	7.1	52.1	32.0	22.4	9	96-371	15.5	355.7	68.7
3.A									347.6	65.6
4.A	88.0	2.1	58.0	28.2	52.4	6	99-369	3.1	353.7	65.1
5.A									356.8	67.6
6.A	82.2	8.7	50.8	22.3	31.2	9	0-339	18.4	335.5	62.6
Site UP25 P_2l (S = 239.8°, D = 89.0°NW)										
1.A									36.5	78.2
2.A	80.1	-5.5	66.0	20.1	27.4	5	99-291	4.0	4.3	66.7
3.A									10.0	65.5
4.A	72.6	-4.4	64.5	12.7	50.0	8	86-365	15.8	351.2	59.6
5.A	91.8	5.0	54.5	32.0	29.7	9	0-366	9.9	355.4	52.1
6.A	77.0	-7.3	67.7	16.9	43.6	7	90-301	5.0	358.0	56.0
Site UP24 P_2l (S = 239.8°, D = 89.0°NW)										
1.A									7.1	68.9
2.A	80.5	-4.4	64.9	20.6	39.0	8	86-339	11.8	1.6	53.9
3.A	96.2	10.0	48.2	36.0	45.3	6	88-256	13.1	351.1	63.1
4.A	80.5	2.6	57.4	20.7	61.9	8	178-368	9.5	344.1	66.9
5.A	74.6	5.1	54.8	14.8	55.7	8	94-345	9.9	333.7	64.5
6.A										
Site UP23 P_2l (S = 239.8°, D = 89.0°NW)										
1.A	63.0	-21.7	81.6	2.6	38.9	7	75-326	5.3	10.4	56.3
2.A	56.2	-26.5	86.3	-3.7	38.2	9	74-384	10.4	8.8	54.4
3.A	66.4	-15.8	75.8	6.1	49.4	9	73-382	6.3	6.9	60.5
4.A	68.5	-16.4	76.5	8.1	46.1	10	73-406	14.1	5.5	58.3
5.A	63.9	-14.9	74.8	3.7	30.7	6	73-290	3.2	24.8	71.1
6.A	65.9	-26.5	86.5	5.0	32.6	8	150-372	7.4	14.3	53.9
Quarry Section (Latitude, 43.77°N; Longitude, 87.58°E)										
Site UP31 P_2h (S = 34.6°, D = 50.7°SE)										
1.A	225.8	19.8	205.6	20.9	22.9	11	0-415	17.3	30.4	73.3
2.A	223.6	25.5	199.3	22.5	28.1	11	0-419	14.1	5.6	62.3
3.A	222.8	18.3	205.1	17.7	152.2	11	173-445	9.4	19.4	69.6
4.A	239.1	22.7	210.8	32.7	53.7	10	178-446	18.3	26.0	63.9
5.A	189.0	14.7	186.7	-9.4	58.6	6	0-272	14.7	2.6	64.8
6.A									350.4	71.6
7.A									4.3	64.5
Site UP30 P_2h (S = 34.6°, D = 50.7°SE)										
1.A	205.4	19.6	193.7	5.5	112.9	12	88-475	7.8	13.6	68.2
2.A	192.3	20.7	184.7	-2.9	29.4	9	90-423	11.9	1.9	66.5
3.A	190.4	19.2	184.5	-5.2	44.9	8	94-345	14.8	10.5	71.3
4.A	203.3	21.9	190.7	5.5	36.7	10	96-425	12.6	12.0	65.1
5.A	202.9	19.1	192.6	3.4	103.8	12	98-477	11.3	359.9	65.3
6.A	201.4	25.5	186.8	6.5	63.5	10	99-445	11.3	18.1	61.6
Site UP29 P_2h (S = 34.6°, D = 50.7°SE)										
1.A	223.1	15.4	207.7	16.2	97.9	8	100-385	6.5	2.9	70.3
2.A	250.3	16.6	224.3	37.9	78.6	7	209-384	10.1	10.4	61.9
3.A	224.0	16.4	207.4	17.5	62.2	5	209-324	2.1	5.8	69.3
4.A	223.4	28.9	196.1	24.2	91.1	7	98-380	3.3	20.0	59.3
5.A	228.8	26.1	201.6	26.7	85.7	7	98-347	4.5	15.3	59.6
6.A	222.4	22.4	201.4	19.8	60.8	6	98-317	15.3	37.2	80.7
7.A	217.7	20.2	200.5	15.0	56.9	7	0-309	5.3	349.2	60.9
8.A	218.0	22.4	198.9	16.5	60.1	8	0-322	5.2	22.5	67.4
9.A.										
Site UP27 P_2h (S = 34.6°, D = 50.7°SE)										
1.A	251.9	11.5	230.5	35.9	83.7	8	100-441	12.0	23.8	72.8
2.A	226.3	25.8	200.5	24.6	97.0	10	0-440	16.4	354.5	49.1
3.A	229.4	25.5	202.5	26.8	86.5	7	99-438	11.3	21.3	61.7
4.A	240.7	19.3	215.1	32.0	66.6	8	98-406	14.2	27.7	67.4
5.A	243.1	0.8	232.9	22.2	76.9	9	0-432	26.2	347.8	64.4
6.A	238.2	5.5	225.7	21.7	66.3	7	98-398	17.0	2.7	80.6
7.A	225.6	17.2	207.7	19.2	83.5	8	97-388	23.6	354.2	63.1

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TABLE 2. (continued)

Specimen	A	P	A_c	P_c	Arc	N	Range	MAD	VRM_D	VRM_I
Hongshan Section (Latitude, 43.80°N; Longitude, 87.62°E)										
Site UP01 P_2l (S = 242.6°, D = 73.0°NW)										
1.A									18.2	65.3
2.A	108.5	10.0	66.0	46.6	6.3	5	209-325	6.4	3.3	54.4
3.A									2.3	62.2
4.A									352.4	58.5
5.A									326.0	77.4
6.A										
7.A										
Site UP02 P_2l (S = 242.6°, D = 73.0°NW)										
1.A	95.8	4.8	68.0	33.1	10.9	6	251-387	9.8	359.7	51.4
2.A	82.1	-5.7	74.2	16.8	25.7	7	226-391	10.6	2.3	48.9
3.A									1.7	57.7
4.A	104.8	5.1	71.1	41.7	31.7	11	90-423	11.6	7.6	54.2
5.A	82.8	-7.0	75.7	17.0	19.3	10	94-396	7.6	4.4	58.6
Site UP03 P_2l (S = 240.1°, D = 69.9°NW)										
1.A	79.0	-13.6	79.8	12.4	15.1	9	237-424	4.1	12.1	62.2
2.A	107.2	5.5	73.4	45.9	6.4	8	242-394	11.3	8.4	62.1
3.A	93.8	-6.3	79.6	28.7	23.5	10	83-387	14.2	14.1	58.3
4.A	65.3	-18.9	79.6	-1.8	26.4	10	86-391	22.8	9.7	59.5
5.A	82.0	-12.0	79.6	15.7	18.7	10	88-393	10.0	9.4	54.5
6.A	91.0	-4.9	76.8	26.8	12.9	10	90-394	12.2	10.4	62.4
Site UP04 P_2l (S = 240.1°, D = 69.9°NW)										
1.A	103.4	-3.5	82.0	38.5	16.6	9	224-424	18.3	17.9	59.6
2.A	92.9	-9.5	82.3	26.4	66.4	9	96-397	15.5	18.6	58.3
3.A	44.0	-16.9	71.3	-20.4	170.5	12	0-424	17.4	8.0	69.4
4.A	105.4	-1.5	81.0	41.2	54.5	7	242-369	5.3	18.9	65.4
5.A	97.5	-6.0	81.3	32.1	99.8	9	160-387	16.2	19.2	58.2
6.A	31.5	-25.8	78.3	-33.7	53.2	9	86-391	12.2	23.6	64.0
Site UP05 P_2l (S = 245.6°, D = 70.0°NW)										
1.A									14.5	69.7
2.A	134.7	21.1	58.7	70.4	13.6	6	301-423	8.9	13.8	58.4
3.A	10.5	-33.8	97.0	-56.2	19.3	9	224-424	9.0	359.3	58.1
4.A	90.8	-4.0	78.7	22.0	9.8	10	96-397	5.7	8.6	62.8
5.A	83.9	-9.7	81.4	13.5	11.6	9	237-424	12.8	11.0	59.2
6.A	114.9	8.6	75.8	49.1	9.9	11	99-419	11.5	5.9	66.6
Site UP08 P_2l (S = 244.6°, D = 68.0°N)										
1.A	113.5	5.1	81.5	46.8	23.0	7	150-347	4.5	15.7	59.3
2.A	87.7	-16.8	89.5	13.9	33.7	6	206-344	6.3	13.3	46.9
3.A	116.7	5.8	82.8	50.0	37.8	8	72-336	14.0	11.7	58.0
4.A	121.7	10.4	79.5	56.4	20.3	8	150-348	10.4	16.9	51.6
5.A	80.0	-11.9	81.6	9.4	11.0	9	72-363	10.7	6.4	53.5
6.A	100.4	-1.2	81.0	32.3	17.3	9	72-373	13.9	12.6	61.1
East Section (Latitude, 43.78°N; Longitude, 87.75°E)										
Site UP32 P_2j (S = 239.5°, D = 75.5°NW)										
1.A	69.4	-10.6	72.3	6.8	43.8	14	75-523	12.0	11.8	60.5
2.A	178.5	-16.3	193.5	47.9	45.2	13	150-523	19.2	52.8	49.2
3.A	73.3	-8.4	71.3	11.1	58.8	14	73-522	8.9	2.8	58.9
4.A	68.5	-35.5	96.0	-1.3	55.2	14	73-520	16.8	12.0	62.0
5.A	61.5	-31.6	90.7	-5.9	66.1	15	0-515	12.0	8.3	57.3
6.A	72.1	-25.8	87.7	4.7	30.7	14	73-508	9.7	8.4	49.9

A is in situ azimuth of demagnetization plane pole in degrees; P is in situ plunge of demagnetization plane pole in degrees; A_c is corrected azimuth of demagnetization plane pole in degrees; P_c is corrected plunge of demagnetization plane pole in degrees; Arc is great circle arc length in degrees subtended by the magnetization vector in response to thermal demagnetization; N is number of magnetic vectors (including the origin) used in least squares planar fits; Range is range of thermal demagnetization steps used in least squares planar fits; MAD is Maximum Angular Deviation defined by *Kirschvink* [1980] for least squares planar fits; VRM_D is viscous remanent magnetization in situ declination in degrees; VRM_I is viscous remanent magnetization in situ inclination in degrees. Each section is listed separately with sites arranged in stratigraphic order. Locations and strikes and dips are given for each section and site, respectively.

significant at p = 0.05. The age of characteristic magnetization therefore predates Neogene folding. A mean direction of magnetization for the four direct observation sites is D = 159.3°, I = -64.2°, R = 3.9174, k = 36, a_{95} = 15.5°.

Demagnetization planes for each of the 129 specimens are classified according to the great circle arc length subtended by the magnetization vector in response to thermal demagnetization (Tables 2 and 3). Longer arc lengths should produce better, least squares planar fits to the demagnetization data. Five populations were created: 11 specimens with arc lengths greater than 100°; 13 with arc lengths greater than 75° but less than 100°; 28 with arc lengths greater than 50° but less

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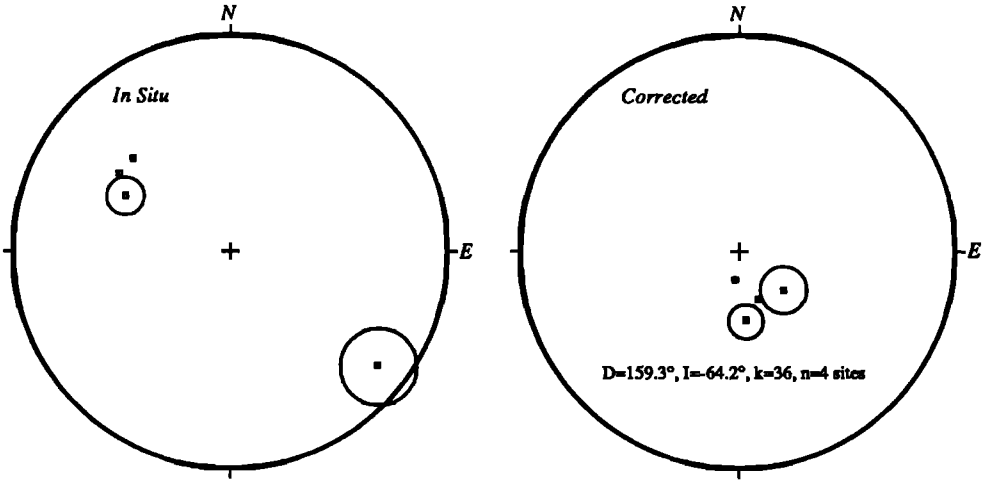


Fig. 12. Equal-angle stereographic projections of the four direct observation site mean directions of magnetization before (left) and after (right) corrections for tilt of beds. Open symbols indicate projections onto the upper hemisphere. Circles are intervals of 95% confidence about the mean directions. Mean directions for sites 8UP33 and 8UP34 have confidence intervals less than 2.5° and thus are plotted without. An overall, tilt-corrected mean is listed.

TABLE 3. Remagnetization Circle Partitions and Mean Directions of Magnetization

Arc	D	I	N	R	k	a_{95}	Iter
> 100°	186.1	-67.4	11	10.5956	11	15.4	9
75° < < 100°	155.2	-44.5	13	12.6958	18	10.7	17
50° < < 75°	153.3	-52.4	28	26.8075	11	8.7	15
25° < < 50°	154.9	-32.7	47	44.8766	11	6.7	22
< 25°	153.6	-20.5	30	29.1791	17	6.6	16

Demagnetization planes are classified according to the great circle arc length subtended by the magnetization vector in response to thermal demagnetization. Mean directions are from the maximum likelihood analysis described in text. Arc is great circle arc length; D is tectonically corrected declination in degrees; I is tectonically corrected inclination in degrees; N is number of specimens; R is resultant vector; k is precision parameter estimate; a_{95} is radius of 95% confidence in degrees; Iter is number of iterations required to converge upon a solution.

than 75°; 47 with arc lengths greater than 25° but less than 50°; and 30 with arc lengths less than 25°.

We used the analysis of *McFadden and McElhinny* [1988] to combine the remagnetization circles and direct observations. Given M direct observations and N great circles, a maximum likelihood analysis is performed using an iterative procedure. The positions of N variable direction unit vectors (a unit vector is confined to lie within the plane of its great circle) are determined so that the length of the vector resultant of all (M + N) unit vectors is a maximum. The iterative procedure applied to just the tectonically corrected remagnetization circles (M = 0) revealed a dependence of the mean direction on grouping according to great circle arc length (Table 3).

Most of the 30 specimens with arc lengths less than 25° showed just a slight shallowing of inclination as the remanent magnetization moved from the present-day field direction with thermal demagnetization, regardless of structural setting. In situ poles to least squares demagnetization planes cluster about a horizontal plunge and an azimuth of either 90° or 270° (Figure 13). The high-T_{ub} component of magnetization could hardly be influencing such great circle trajectories. Therefore exclusion of these data, as well as the 47 demagnetization planes defined by arc lengths greater than 25° but less then 50°, is warranted. A mean direction of magnetization for the remaining 52 tectonically corrected demagnetization planes defined by arc lengths greater than 50° is D = 160.8°, I = -56.3°, R = 49.8675, k = 12, a_{95} = 6.0°, *iterations* = 13.

A final mean direction combining the M = 26 direct observations with the N = 52 demagnetization planes is given as D = 163.0°, I = -61.9°, R = 74.9838, k = 17, a_{95} = 4.0°, n = 78 specimens, *iterations* = 5. Sector constraints, in the form of an arc of the great circle along which the estimate of the actual direction must lie, were not used [*McFadden and McElhinny*, 1988].

Discussion

The mean direction of the high-T_{ub} magnetization obtained from the combined analysis of direct observations and remagnetization circles could be either a primary Permian magnetization or a postdepositional but prefolding remagnetization. The former interpretation is preferred for the following two reasons. First, the observation of only reversed directions of magnetization is consistent with the mid-Carboniferous to Late Permian Kiaman reversed interval for the geomagnetic field [*Harland et al.*, 1982]. Second, vitrinite reflectance values of 0.40-0.59 and tar and oil observed in fractures testify to the thermal immaturity of the exposed Lucaogou Formation mudstones with respect to both oil generation [*Graham et al.*, 1990] and possible remagnetization processes.

Alternatively, the characteristic magnetization could be a low-temperature chemical remanent magnetization acquired during a prefolding remagnetization event. For example, an authigenic growth of magnetite could have been in response to

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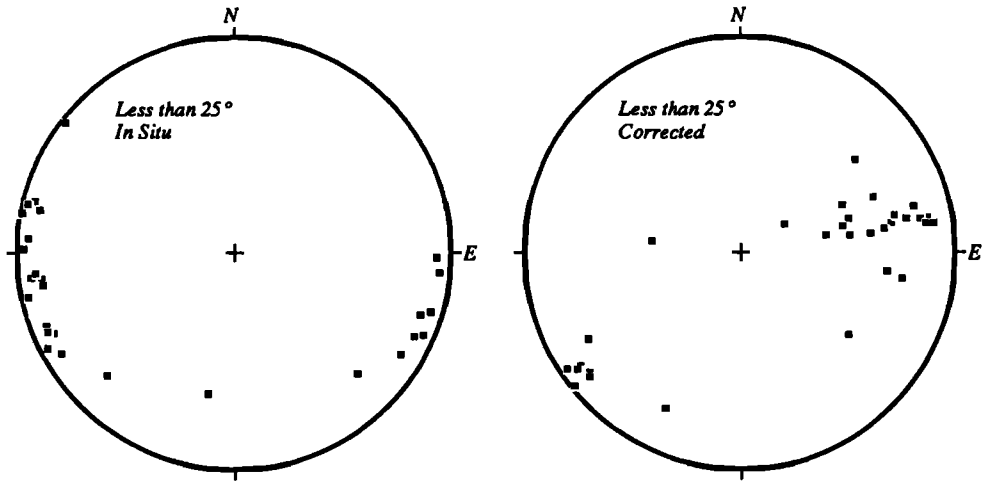


Fig. 13. Equal-area stereographic projections of poles to least squares demagnetization planes for 30 specimens, both before (left) and after (right) corrections for tilt of beds. Less than 25° of great circle arc length was subtended by the magnetization vector in response to thermal demagnetization for each of these 30. Solid symbols indicate projections onto the lower hemisphere.

migration of fluids into the Permian foreland basin of Junggar from the Tian Shan orogenic front [cf. *McCabe et al.*, 1983; *Jackson et al.*, 1988]. Such a remagnetization would have occurred soon after deposition; therefore its direction would not differ significantly from the primary direction of magnetization, and the original interpretation is equally valid. In theory, an early Tertiary remagnetization is also possible. Much of the outcrop relief around Urumqi appears to be a consequence of compression related to Tertiary convergence between the Indian and Eurasian plates; however, there is no other supporting evidence for a Tertiary remagnetization.

The strong, low-T_{ub} normal direction is unquestionably a postfolding, Brunhes viscous component of magnetization.

On the basis of the observations mentioned above, we consider the reversed magnetization to be a primary Permian depositional remanence. A comparison of paleomagnetic results obtained from Late Carboniferous volcanics and sediments sampled from the Liushugou, Qijiagou, and Aoertu formations exposed east of Urumqi [*Li et al.*, 1991], late Paleozoic volcanic flows, dikes, and sediments near Karamay in northwest Junggar [*Li et al.*, 1991; *Li et al.*, 1989], and Upper Permian clastics near Tianchi northeast of Urumqi [*Nie et al.*, 1989] reveal similar directions of magnetization (Figure 14). Consistently steep inclinations indicate little apparent polar wander and argue for a cohesive Junggar "block" since Carboniferous times.

Competing hypotheses concerning the tectonic origin and development of the Junggar basin (critically evaluated by *Carroll et al.* [1990]) are based upon opinions regarding the composition of basement material beneath the basin fill. No lower Paleozoic or older crystalline rocks are exposed in the area immediately surrounding the Junggar basin. *Carroll et al.* [1990], in light of the stratigraphic and sedimentologic character of Carboniferous and Permian sequences, favored either a remnant, "trapped" ocean basin or a back-arc basin tectonic setting for the Junggar region between emergent volcanic arcs during late Paleozoic times. Upper Permian sediments in the South Junggar basin were deposited in a collisional, flexurally subsiding foreland basin, consistent with the low levels of thermal maturation attained by the exposed Permian lacustrine organic-rich mudstones. A marginal ocean basin setting for Junggar would predict little apparent polar wander, as well as little if any latitudinal motion relative to Siberia.

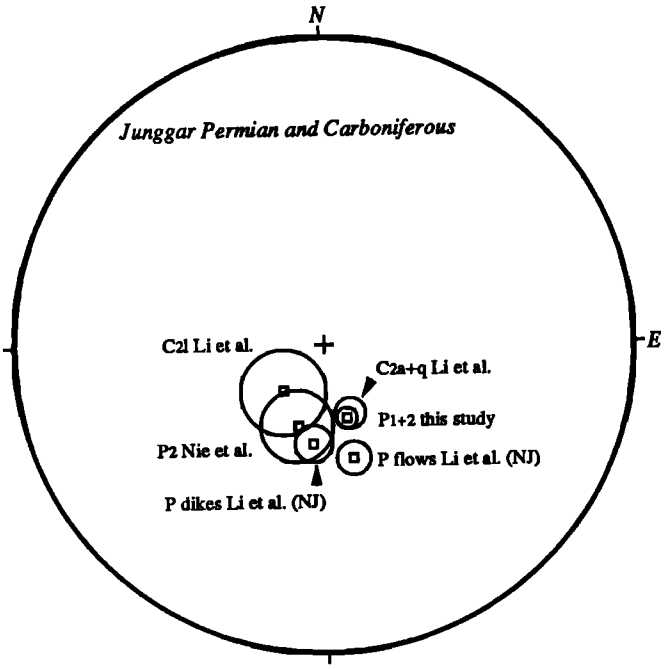


Fig. 14. Equal-angle stereographic projection of Permian and Carboniferous mean directions of magnetization, reported from the South Junggar and North Junggar (NJ) regions. Open symbols indicate projections onto the upper hemisphere. Circles are intervals of 95% confidence about the mean directions. Full references are given in the text.

The steep mean inclination of -61.9° ± 4.0° preserved within the lacustrine mudstones places the Late Permian foreland basin of South Junggar in latitudes around 43°. The northern Tian Shan calc-alkaline volcanic arc to the immediate south may have marked not only the southern active boundary of Angaraland during the Permian but also the northern extent of paleo-Tethyan oceanic lithosphere. Permian plants in the Junggar basin are mainly Angaran flora, identical to species found in the Kuznetsk basin of the USSR [*Hu*, 1985]. A few Cathaysian flora are also present; however, the major biogeographical boundary between Angaran flora to the north and Cathaysian flora to the south coincides with the structural trend of the Tian Shan [*Hu*, 1985].

On the basis of the superposition of Lower Permian clastics, the Upper Permian lacustrine organic-rich mudstones, and a prograding Lower Triassic fluvial clastic sequence in the Junggar basin, *Watson et al.* [1987] proposed a primary

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tectonic control on lacustrine rock deposition in a deep lake setting under humid conditions. Assuming significant topographic rise of the ancestral Tian Shan, climatic conditions at latitudes of 43° during the Late Permian (latitude at which deposition may have taken place) must have been drastically different than the semiarid conditions of present-day Junggar. An open ocean south of the ancestral Tian Shan may have created the required moderate, maritime climate.

In contrast to Junggar, rapid apparent polar wander and northward translation from the Devonian through to the Permian for Tarim suggest that the terrane was attached to paleo-Tethyan oceanic lithosphere subducting northward beneath the Tian Shan volcanic arc [*Li*, 1988]. Timing of volcanism supports active subduction beneath the Bogda Shan into Permian times [*Carroll et al.*, 1990]. A Permian collision of Tarim with the northern Tian Shan volcanic arc is suggested by paleomagnetic directions from Lower Permian sediments and volcanics, which impose latitudinal proximity of northwest Tarim with South Junggar [*Sharps et al.*, 1989], and by a possible Permian remagnetization of Lower and Middle Carboniferous marine carbonates exposed on northern Tarim as perhaps a consequence of collisional deformation [*Sharps*, 1990]. A major change in plate motion, as indicated by a cusp in the apparent polar wander path for Tarim, occurred in the Triassic. This change is perhaps a manifestation of the continent/arc collision of Tarim with the Tian Shan and Junggar.

In conclusion, we feel that our paleomagnetic data as well as the published sedimentological data support a marginal oceanic basin setting for Junggar, south of Carboniferous and Permian Laurasia. Rifting of northernmost Gondwana created terranes such as Tarim, which drifted from south to north as Tethyan oceanic lithosphere subducted through stationary trenches [*Sengör*, 1984]. The Permian accretion of Tarim to the Tian Shan volcanic arc led to the rise of the ancestral Tian Shan, the transition from marine to nonmarine facies sedimentation in both the Tarim and Junggar basins, and the development of the collisional foreland basin of South Junggar into which the lacustrine sediments deposited. Junggar has experienced little if any apparent polar wander since Carboniferous times, and its latitudinal position relative to Siberia [cf. *Khramov et al.*, 1985] has remained unchanged.

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REFERENCES

Bureau of Geology and Mineral Resources of Xinjiang Uygur Autonomous Region, Geological map of the Urumqi area, scale 1:200,000, Geological Publishing House, Beijing, China, 1983.

Carroll, A. R., Y. Liang, S. A. Graham, X. Xiao, M. S. Hendrix, J. Chu, and C. L. McKnight, Junggar Basin, Northwest China: Trapped late Paleozoic ocean, *Tectonophysics*, *181*, 1-14, 1990.

Dekkers, M. J., Magnetic properties of natural goethite-I. Grain-size dependence of some low- and high-field related rockmagnetic parameters measured at room temperature, *Geophys. J.*, *97*, 323-340, 1989*a*.

Dekkers, M. J., Magnetic properties of natural goethite-II. TRM behaviour during thermal and alternating field demagnetization and low-temperature treatment, *Geophys. J.*, *97*, 341-355, 1989*b*.

Feng, Y., R. G. Coleman, G. Tilton, and X. Xiao, Tectonic evolution of the West Junggar region, Xinjiang, China, *Tectonics*, *8*, 729-752, 1989.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Res.*, *54*, 131-167, 1949.

Graham, S. A., S. Brassell, A. R. Carroll, X. Xiao, G. Demaison, C. L. McKnight, Y. Liang, J. Chu, and M. S. Hendrix, Characteristics of selected petroleum-source rocks, Xinjiang Uygur Autonomous Region, Northwest China, *AAPG Bull.*, *74*, 493-512, 1990.

Harland, W. B., A. V. Cox, P. G. Llewelyn, C. A. G. Pickton, A. G. Smith, and R. Walter, *A Geologic Time Scale*, 131 pp., Cambridge University Press, New York, 1982.

Hsü, K. J., Relict back-arc basins: principles of recognition and possible new examples from China, in *New Perspectives in Basin Analysis*, edited by K. L. Kleinspehn and C. Paola, pp. 245-263, Springer-Verlag, New York, 1988.

Hu, Y., Characteristics of the Permian floras in the western part of China (in Chinese with English abstract), *Reg. Geol. China*, *12*, 99-108, 1985.

Jackson, M., C. McCabe, M. M. Ballard, and R. Van der Voo, Magnetite authigenesis and diagenetic paleotemperatures across the northern Appalachian basin, *Geology*, *16*, 592-595, 1988.

Khramov, A. N., E. L. Gurevich, R. A. Komissarova, E. P. Osipova, S. A. Pisarevsky, V. P. Rodionov, and I. P. Slautsitais, Paleomagnetism, microplates and Siberian plate consolidation, *J. Geodyn.*, *2*, 127-139, 1985.

Kirschvink, J. L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc.*, *62*, 699-718, 1980.

Kwon, S.-T., G. R. Tilton, R. G. Coleman, and Y. Feng, Isotopic studies bearing on the tectonics of the West Junggar region, Xinjiang, China, *Tectonics*, *8*, 719-727, 1989.

Li, Y.-P., Paleomagnetism of western China and the southern Sierra Nevada, Ph.D. thesis, 234 pp., Stanford Univ., Stanford, Calif., 1988.

Li, Y.-P., R. Sharps, M. McWilliams, A. Nur, Y.-A. Li, Q. Li, and W. Zhang, Paleomagnetic results from late Paleozoic dikes from the northwestern Junggar block, northwestern China, *Earth Planet. Sci. Lett.*, *94*, 123-130, 1989.

Li, Y.-P., R. Sharps, M. McWilliams, Y.-A. Li, Q. Li, and W. Zhang, Late Paleozoic paleomagnetic results from the Junggar block, northwestern China, *J. Geophys. Res.*, *96*, 16047-16060, 1991.

McCabe, C., R. Van der Voo, D. R. Peacor, C. R. Scotese, and R. Freeman, Diagenetic magnetite carries ancient yet secondary remanence in some Paleozoic sedimentary carbonates, *Geology*, *11*, 221-223, 1983.

McElhinny, M. W., Statistical significance of the fold test in palaeomagnetism, *Geophys. J. R. Astron. Soc.*, *8*, 338-340, 1964.

McFadden, P. L., and M. W. McElhinny, The combined analysis of remagnetization circles and direct observations in palaeomagnetism, *Earth Plant. Sci. Lett.*, *87*, 161-172, 1988.

Nie, S., D. B. Rowley, R. Van der Voo, and M. Li, Late Paleozoic paleomagnetic results from Northwest China and Permian Asian plate reconstructions, *Eos Trans. AGU*, *70*, 1071, 1989.

Ren, J., C. Jiang, Z. Zhang, and D. Qin, *Geotectonic Evolution of China*, 203 pp., Springer-Verlag, New York, 1987.

Sengör, A. M. C., The Cimmeride Orogenic System and the tectonics of Eurasia, *Spec. Pap. Geol. Soc. Am.*, *195*, 82 pp., 1984.

Sharps, R., Paleomagnetism and tectonics of Northwest China, Ph.D. thesis, 250 pp., Stanford Univ., Stanford, Calif., 1990.

Sharps, R., M. McWilliams, Y.-P. Li, A. Cox, Z. Zhang, Y. Zhai, Z. Gao, Y.-A. Li, and Q. Li, Lower Permian paleomagnetism of the Tarim block, northwestern China, *Earth Planet. Sci. Lett.*, *92*, 275-291, 1989.

Watson, M. P., A. B. Hayward, D. N. Parkinson, and Zh. M. Zhang, Plate tectonic history, basin development and petroleum source rock deposition onshore China, *Mar. Pet. Geol.*, *4*, 205-225, 1987.

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